

WORK EXPERIENCE

WANDERLOG

San Francisco, CA

Full Stack Software Engineer

May 2025– Present

- Developed a new user profile page in **React Native (TypeScript)**, with a **REST API** and an **Express** backend, adding social features and a modern UI for millions of users of a travel app on the App Store.
- Optimized web performance and Core Web Vitals by reducing CLS by 50% and cutting LCP from 5.1s on average to 1.6s on average by enhancing server-side rendering, improving load speed and stability on low-end devices.
- Led development for the web interface for the app's trip journaling system using **React**, integrating with existing mobile editing functionality, building new web editing functionality and delivering a responsive view based on the size of the user's device for over 300,000 monthly users.

MICRONOVA LLC (Kickstarter-backed **open source** home audio startup)

Waterford, MI

Software Engineering Intern

May 2024 – January 2025

- Enabled robust automation options and enhanced user experience for thousands of global users by extending a Home Assistant integration with a **Python** backend and a **TypeScript** frontend, interfaced during development via **Docker**.
- Implemented highly requested functionality from a long-standing GitHub issue by developing a local media and file browser, which uses a **ReactJS** web frontend that communicates with a **Python** backend via a **REST API** powered by **FastAPI**.
- Expanded smart home capabilities for customers, enabling seamless home audio control, by engineering **embedded firmware** in **C++** for an **ESP32**-powered wall panel and lightswitch, an upcoming accessory to the home audio system.

USAA

San Antonio, TX

Software Engineering Intern

May 2023 – Aug 2023

- Optimized data retrieval and efficiency for millions of USAA members by contributing towards the development of a **REST API** in an **Agile** environment, built using **Java** with **Spring Boot** and **Gradle**, and interfacing with a **SQL** database.
- Streamlined project migration to newer internal frameworks by reducing configuration file overhead by 50% in **YAML**.
- Ensured a more reliable system, decreasing critical and high severity vulnerabilities by 90%, by upgrading libraries and packages.

PROJECTS

YOUTUBE CHANNEL

Jul 2017 - Present

- Gained a dedicated online following with over 20,000 subscribers and 2 million video views by producing over 50 animated educational videos on programming focused topics.
- Researched over 30 different programming languages, wrote example programs in them ranging from popular algorithms to games, and presented the programs in detail while explaining how the language works.

A REAL TIME STRATEGY VIDEO GAME

Feb 2024 - Apr 2024

- Developed a large and complex video game in **Unity** within 6 weeks consisting of over 150 **C#** scripts and 100+ media assets, managed using **Jira** and **Git**, which received 5th place out of 21 at an end-of-year showcase attended by hundreds of players.

EDUCATION

UNIVERSITY OF MICHIGAN - ANN ARBOR

Ann Arbor, MI

Masters of Engineering in Computer Science

May 2025

Relevant Coursework: Advanced Operating Systems, Advanced Compilers, Parallel Computing

UNIVERSITY OF MICHIGAN - ANN ARBOR

Ann Arbor, MI

Bachelors of Engineering in Computer Science, Minor in German

May 2024

GPA: 3.74/4.0

Relevant Coursework: Web Systems, Intro to AI, Data Structures & Algorithms, Intro to OS, Compiler Construction

ACTIVITIES

UM::AUTONOMY ROBOBOAT TEAM

Ann Arbor, MI

Task Planning Software Lead

Sep 2020 – Mar 2024

- Led the task planning software team to calculate effective waypoints for an autonomous water vehicle, enabling the team to place 4th at the 2024 RoboBoat competition, by developing **multi-threaded Python** and **C++** applications, running in **parallel** on **ROS** with other systems.

SKILLS

Programming Languages: Python, C, C++, C#, Java, JavaScript, TypeScript, JSON, YAML, SQL, Rust, PHP, Lua

Frameworks and Libraries: Flask, FastAPI, React, React Native, Spring Boot, GraphQL, Express.js, Tailwind

Tools: Cursor, Gradle, FFmpeg, Unity, Git, GitHub, GitLab, Docker, Home Assistant, Jira

Operating Systems: Windows, Linux, macOS